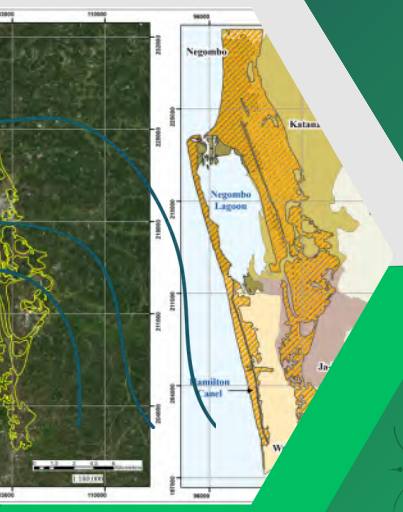
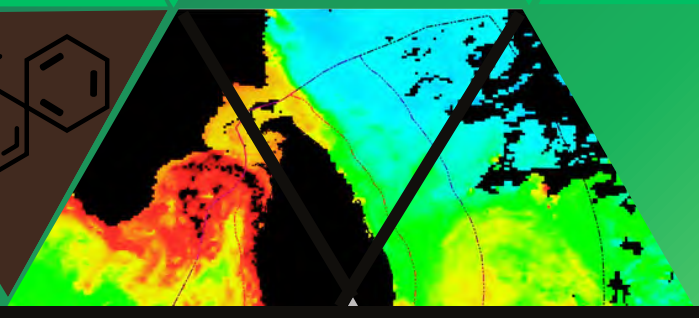
POSTGRADUATE INSTITUTE OF SCIENCE
UNIVERSITY OF PERADENIYA

POSTGRADUATE RESEARCH HIGHLIGHTS







PGIS RESCON 2025

Ph.D. Research

M.Phil. Research

M.Sc. Research

VOLUME 10

POSTGRADUATE INSTITUTE OF SCIENCE
UNIVERSITY OF PERADENIYA
SRI LANKA

POSTGRADUATE
RESEARCH HIGHLIGHTS 2025

RESCON 2025

PGIS POSTGRADUATE RESEARCH HIGHLIGHTS 2025

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MESSAGE FROM THE DIRECTOR



It is with great pride and academic fulfillment that I present the Postgraduate Research Highlights of PGIS 2025, which capture the breadth and depth of postgraduate research across M.Sc., M.Phil., and Ph.D. programmes at the Postgraduate Institute of Science (PGIS), University of Peradeniya. The Postgraduate Research Highlights of PGIS 2025 will be published alongside RESCON 2025, further strengthening the visibility and impact of postgraduate research conducted at the Institute.

Over the years, RESCON has evolved into a premier platform that fosters intellectual exchange, promotes interdisciplinary collaboration, and inspires innovative solutions to challenges of national and global significance. The wide spectrum of research showcased in this volume from advanced computational intelligence, machine learning, and renewable energy technologies to molecular biology, environmental sustainability, plant and zoological sciences, and earth system studies reflects the Institute's unwavering commitment to excellence, relevance, and societal impact.

The diversity of research contributions presented here underscores a defining strength of PGIS: its ability to integrate fundamental scientific inquiry with applied research that addresses contemporary needs. These studies not only advance academic frontiers but also provide valuable insights for sustainable development, public health, natural resource management, and technological innovation.

I extend my sincere appreciation to our postgraduate students and their supervisors for their scholarly rigor, creativity, and perseverance. I also wish to acknowledge the Boards of Study for their continuous guidance, and the Chairperson and Organizing Committee of RESCON 2025 for their dedication and commitment in making this scholarly forum a success.

A special note of appreciation is extended to Dr. Thiloka Ariyasena, Editor-in-Chief; Mr. Prabhath Gunathilake, Managing Editor / PGIS Research Highlights Committee; Dr. Ashwini Amarasinghe, Managing Editor; and the Editorial Committee of the PGIS Research Highlights 2025, for their outstanding efforts in compiling and presenting the diverse research achievements of PGIS.

As PGIS continues to align its research and training with national priorities and global scientific trends, I am confident that the work featured in this volume will serve as a catalyst for future discoveries and innovations, further strengthening our position as a leading centre for postgraduate education and research in Sri Lanka.

Let RESCON 2025 inspire us all to build stronger academic collaborations and to transform scientific knowledge into tangible solutions that benefit our communities, industries, and the environment.

Prof. Nanda Balasooriya
Director, Postgraduate Institute of Science (PGIS)
University of Peradeniya

MESSAGE FROM THE CHAIRPERSON RESCON 2025



It gives me immense pleasure and pride to present the PGIS Research Highlights 2025, published as part of the Postgraduate Institute of Science's (PGIS) mission to enhance research visibility, academic excellence, and industry outreach. This publication brings together a selection of outstanding research discoveries from postgraduate scholars who successfully completed their Ph.D., M.Phil., and M.Sc. degrees in 2024.

Many of these research findings have already appeared in reputable international journals, and several of our scholars have been recognized with prestigious awards for their exceptional contributions to advancing scientific knowledge. The compilation of this volume is a reflection of the dedication, talent, and collaborative spirit of the Research Highlights Committee of RESCON 2025, whose tireless efforts have made this vision a reality.

I extend my heartfelt appreciation to all the postgraduates of the 2024 cohort and their supervisors for their exemplary work, which forms the foundation of this publication. At the most recent convocation, we proudly celebrated the achievements of 301 postgraduates, including 8 Ph.D., 21 M.Phil., and 272 M.Sc./Master's degree recipients each contributing to the vibrant research landscape of the PGIS. Their collective achievements demonstrate the institute's continued commitment to nurturing innovation and excellence across diverse fields of science.

This publication serves not only as a record of scholarly accomplishment but also as an invitation for collaboration, providing a window into the breadth and depth of research carried out at PGIS. Those interested in exploring these topics further are warmly encouraged to connect with the respective authors or their supervisors to foster new academic and industrial partnerships.

I extend my sincere gratitude to the Director of PGIS, Prof. Nanda Balasooriya, for his visionary leadership and unwavering support; to the Editor-in-Chief of Research Highlights of RESCON 2025, Dr. Thiloka Ariyasena for her meticulous editorial guidance; and to the Secretary of RESCON 2025, Dr. Gajaba Ellepola, for his exceptional organizational contribution. My appreciation also goes to the Editorial Committee of the RESCON Research Highlights 2025 for their dedicated and efficient efforts in bringing this publication to life.

Finally, I wish the RESCON 2025 team every success in conducting yet another impactful conference that continues to inspire scientific dialogue, innovation, and collaboration.

Prof. Manawadevi Y.U. Ganehenege
Chairperson
RESCON 2025 & Board of Study in Chemical Sciences
Postgraduate Institute of Science
University of Peradeniya

MESSAGE FROM THE EDITOR-IN-CHIEF



It is with great pleasure that I present the Postgraduate Research Highlights, Volume 10 in parallel to the annual research congress RESCON 2025 of Postgraduate Institute of Science, University of Peradeniya. Postgraduate Research Highlights 2025 is a compilation of selected research works by Ph.D., M.Phil. and M.Sc. (SLQF Level 10) graduates of the Postgraduate Institute of Science who earned their degrees in 2024.

The theme of RESCON 2025, 'Synergy in Science: Uniting Disciplines for a Better Tomorrow' offers a compelling vision for the future. Volume 10, presented herein, reflects the commitment of Postgraduate Institute of Science to advancing high-quality, multidisciplinary research that addresses real-world challenges and contribute to national development. The studies featured in this volume exemplify the multidisciplinary collaborations ranging from physical sciences, environmental and earth sciences, life sciences, health sciences, data sciences, mathematics and science education. Collectively, they demonstrate that innovation emerges when diverse minds and methods intersect.

As the editor-in-chief of Postgraduate Research Highlights 2025, I congratulate all the authors who contributed their research articles. I'm also grateful to their research supervisors, whose guidance has been the foundation for their success. My gratitude also goes to the editorial board of Postgraduate Research Highlights 2025 for their tireless efforts in reviewing the articles and compiling the volume.

I would like to express my sincere gratitude to the Director/PGIS, Prof. Nanda Balasooriya, and the Chairperson/RESCON 2025, Prof. Manawadevi Y.U. Ganehenege, for their continued support and encouragement. My sincere appreciation also goes to the organizing committee of RESCON 2025 for their support in bringing this intellectual forum to completion.

May Postgraduate Research Highlights 2025 continue to inspire our graduates to pursue their research passions, contributing to the advancement of science in Sri Lanka and beyond.

Dr. J.A.Thiloka C. Ariyasena
Editor-in-Chief
PGIS Research Highlights 2025
Postgraduate Institute of Science
University of Peradeniya

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